

Part: **3.3 – METAL ASSEMBLIES**

Chapter: **3360**

Title: **FITTINGS**

Revision: **0.1**

Date: **10/01/2024**

1 INTRODUCTION

1.1 OBJECT AND SCOPE

“Fittings” are structural parts, typically made of metal alloys, aiding to connect two parts or components of the airframe produced independently and mounted in later assembly phases of the manufacturing process.

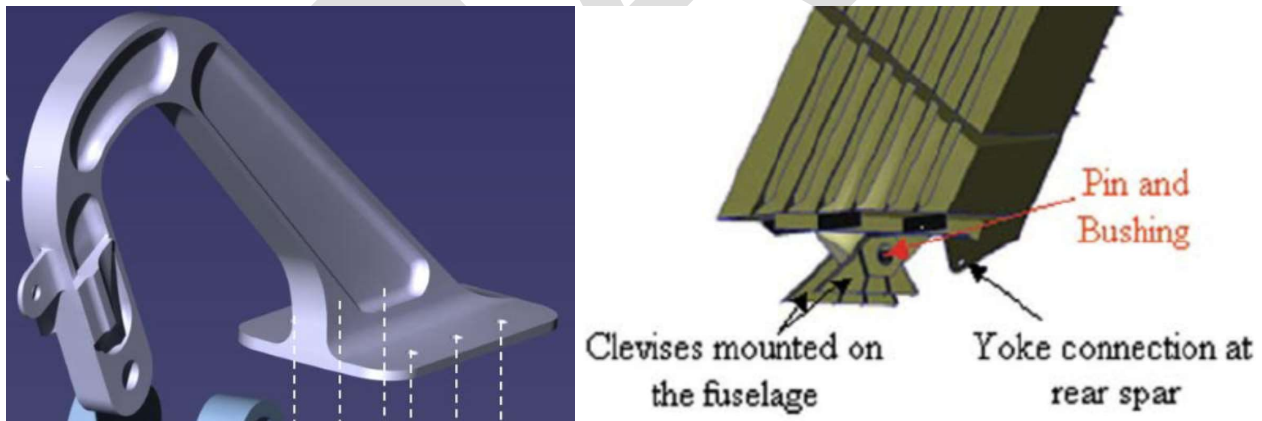


Figure 1. Examples of fittings: “gooseneck” fitting at a hinged door (left) and fitting joining a vertical tail plane with the fuselage (right)

Typical fitting assemblies consist of a metallic machined/forged part bolted to a back-up structure and ended in a lug/clevis or tension end wall feature to be connected, by means of a bolt, to another fitting at the counterpart.

2 METHOD DESCRIPTION

During the initial phases of the airframe design, fittings shall be typically (*) sized using “handbook” methodologies. If the development or certification plan for the airframe includes a dedicated test on the element (which is properly correlated with this analysis) or the fitting is sufficiently similar to a previously certificated fitting, it may be permitted to keep this methodology even until the final certification assessment.

However, some simplifications needed when using handbook methodologies may be questionable. The principal sources of error are the “beam assumption” for the fitting body (if not sufficiently slender), and the stiffness values selected for the rivets attaching the fitting to the back-up structure, especially if tension forces are significant. These uncertainties and the “democratization” of the Finite Elements method has made more popular the use of FEM’s for fittings in order to finally demonstrate compliance with structural requirements, and even as aid in sizing (2.5D or shells models) in intermediate design phases.

(*) New methods and tools as topological optimization would allow to approach the sizing of a fitting with few initial data about its design.

2.1 CONVENTIONAL STRESS & STRENGTH ANALYSIS (HANDBOOK METHODOLOGIES)

For analysis purposes, fittings are normally decomposed in following more simple structural details:

2.1.1 FITTING TERMINAL

Pinned Lugs

This is the most common design for the interface of two structures connected by fittings: the fitting terminal consists of a one or two lugs (in clevis configuration), and a bolt pinned through the lugs complete the connection between the two structures.

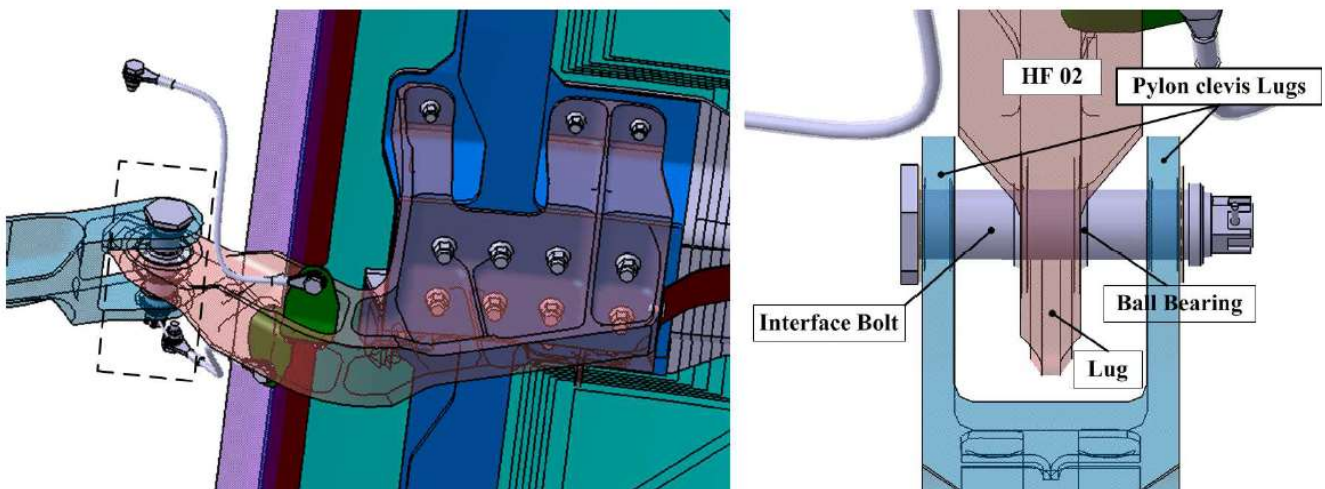


Figure 2. Pinned-Lug clevis attachment at the hinge line of a Fan Cowl of an Airbus aircraft

The strength of the lugs and of the interface pin/bolt shall be calculated according to method described in chapter §3250.

In case there is some tension load at the bolt transferred to some of the lugs, that tension force shall be considered not only in the strength analysis of the bolt (eventually in the ball-bearing if exists) but also in the affected lugs (risk of failure by out-plane shear-bending from sections at the pin to sections at the transition with the fitting body).

Tension end wall

Tension end wall and bolt shall be analysed following the methodology described in chapter §3240.

2.1.2 FITTING BODY

The fitting body is treated as a cantilever beam (straight or curved) and stresses and strengths are calculated using classical methodologies. A limited, sufficient, number of sections are selected, then internal loads calculated (axial and shear forces bending and torsion moments) at those sections.

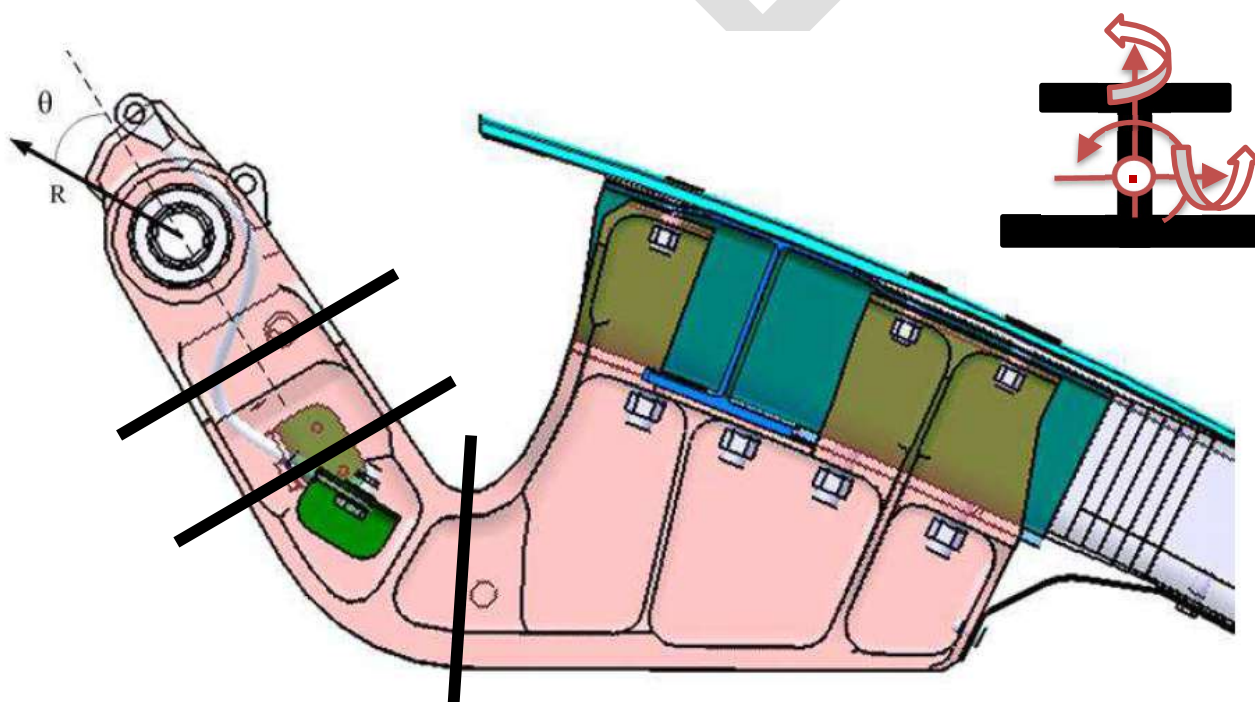


Figure 3. Typical sections that should be analysed in a "slender bodied" fitting

Methods in chapters §3110 and §3111 are used then to obtain the stress distribution at each section of interest. Maximum Von Mises stresses are compared with material yield or ultimate stress as appropriate (see chapter §3102). If a significant part of the section, or of a segment of the section, is in compression stresses, a stability/crippling strength check shall be done also (see chapter §3130).

If the fitting body is compact, not slender, and hardly assimilable to a beam (length of beam of the same order or lower than section dimensions) some engineering judgement should be done to reduce the section considered in analysis to a reasonable/conservative effective size.



Figure 4. Non-slender fitting

2.1.3 BOLTED ATTACHMENT

Rivets loads

Shear and tension forces in the fasteners attaching the fitting to the back-up structure can be calculated by means of the methods described in chapters §3212 to §3214. Estimation of rivets stiffness for the calculation is discussed in chapter §3210. Nevertheless, and especially in eccentric joints with rivets load in tension, the configuration of the fitting body and the back-up structure may have a significant influence in loads paths. Engineering judgement and/or detailed FEM's should be applied to prevent significant error in rivets loads calculations.

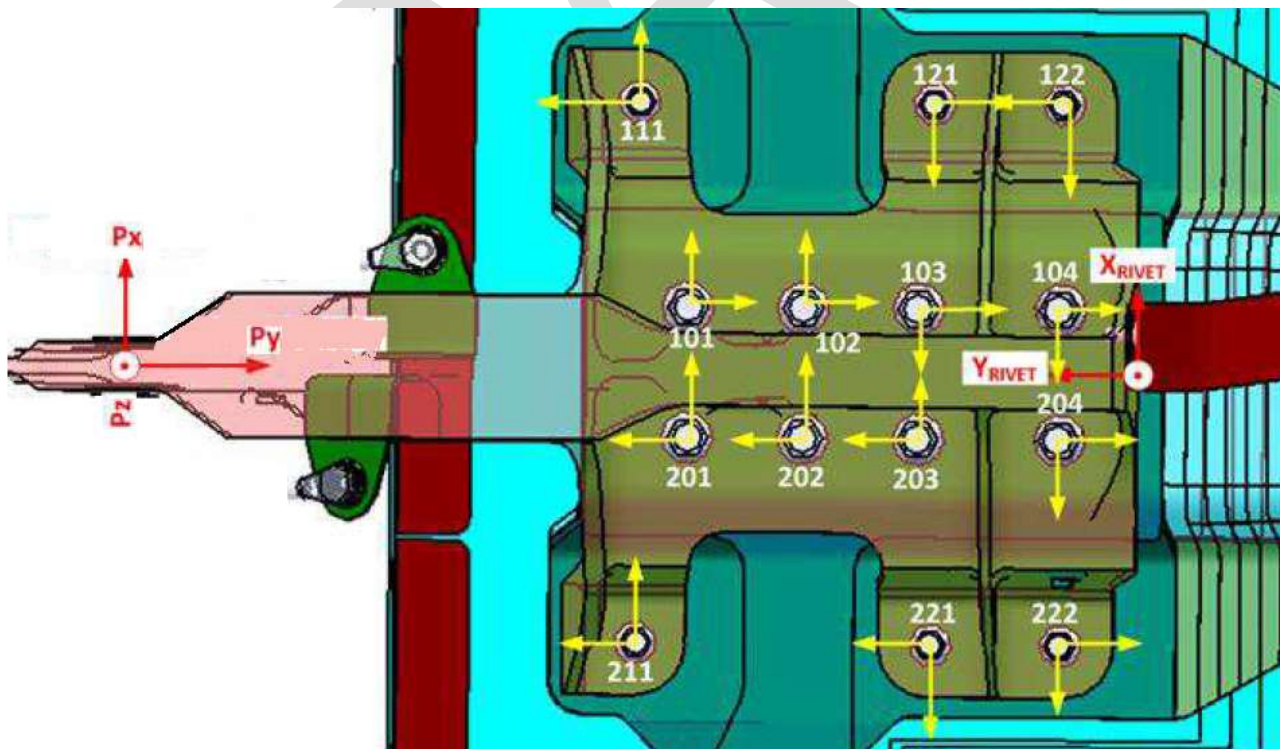


Figure 5. Sketch of reaction forces at the attachment rivets

Strength of joint

Finally, the strength of fasteners and vicinities shall be checked according to methods described in chapters §3220, §3230, or chapter §3620 when fitting is attached to a back-up structure made of composite materials.

With tension loads in rivets, and additional check of the material around the fastener shall be done, as tension fitting, according to chapter §3240.

2.2 DETAILED FINITE ELEMENT MODELS

As discussed in the introduction of this chapter, the following degree of refinement in the stress & strength analysis of metallic fittings is to use Finite Element Models. Details about modelling rules, selection of analysis type (linear elastic or non-linear) and associated considerations, etc... are out of the scope of this chapter. These topics shall be treated in Volume 2, and some guidance can be already found in documents as Ref. 1 or Ref. 2. Following lines provide just a high-level general overview of the possible options for modelizations at this level.

2.2.1 SHELLS MODELS (2.5D FEM'S)

Stress analysis

Many fittings are designed in such a manner that can be seen as plates/shells joined together, in other words, sections of the fitting will resemble thin-walled sections in a conventional beam design. This possibly due to traditional sizing processes based in the handbook stress methodologies like those described in previous paragraphs. Design processes like topological optimization will result in what are frequently called "bionic shapes", which will not match that thin-walled section assumption.

FIGURA PENDIENTE, PREGUNTAR A COMPAS DEL F10X !!!!!!!!!!!!!

Figure 1. Von Mises stress plot in a hinge fitting modelled with shell elements

Thin walled fittings, once its global geometry is known and "only" thickness needs to be tuned, could be modelled used plane stress elements (eg. CQUAD4 elements in MSC NASTRAN software or S4R elements in Simulia ABAQUS). The stress distribution provided by this kind of modelization in the fitting body shall normally be more precise than that obtained with handbook methodologies. Also the loads obtained at rivet positions, especially if the back-up structure to which the fitting is attached is properly represented in the model.

Note this type of modelization cannot handle complex 3D effects, eg. predicting precisely stress fields in fitting material at machining radii or rivets locations.

Strength check

Stresses in fitting body and rivets loads will be the relevant results to recover from the FE analysis. Static margins will be still obtained using the handbook methodologies discussed in previous pages at lug, body (Von Mises in FEM vs material *UTS*), fasteners and their vicinities ("red dots" in material at rivets connectors are normally not taken too seriously).

2.2.2 FULL 3D MODELS

Stress analysis

Full 3D models, where material is modelled using solid elements (hexaedral preferably or tetraedals) are typically used when the design of the fitting is frozen. With sufficient refinement, they could be capable to get precise stress fields at any location of the fitting.

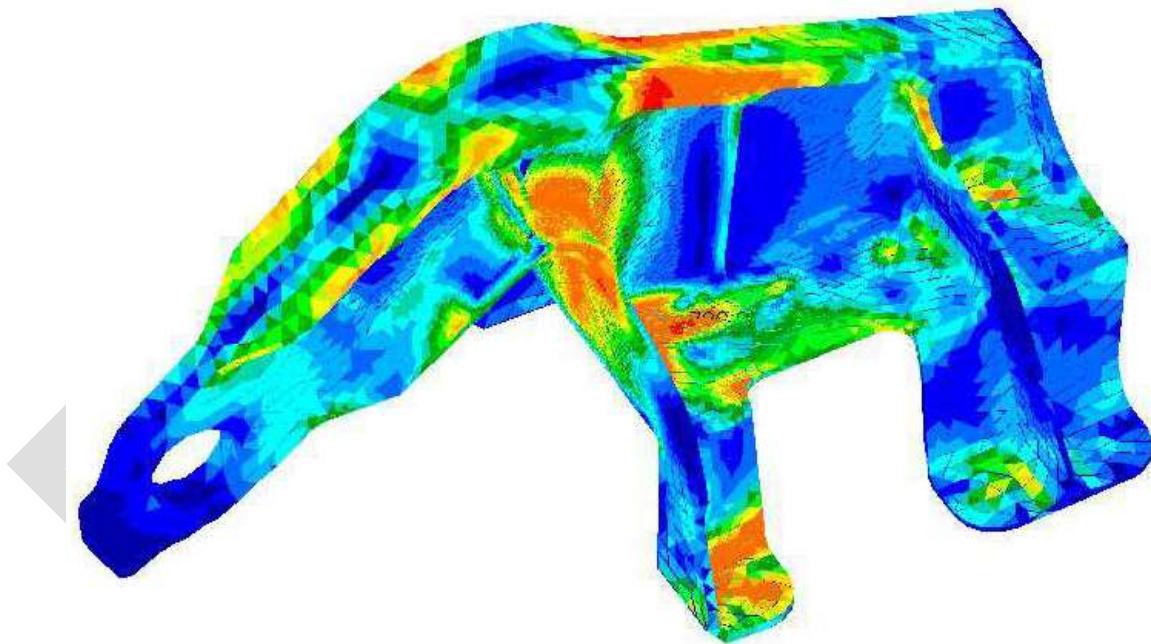


Figure 2. Von Mises stress plot in a hinge fitting modelled with solid tetrahedral elements

Strength check

With sufficient resolution even fasteners could be included in the FEM simulation and their strength, not only that of the fitting body, would be checked also comparing the material *UTS* with the Von Mises stress calculated by the model (non-linear effects including material inelastic behaviour should be accounted).

The strength check of the lug and the bearing failure at rivets position should be still checked using the handbook methodologies previously discussed.

2.3 ADVANCED NUMERICAL SIMULATION

The highest degree of fidelity in predictions of static failure of a metallic part can be obtained combining full 3D models and advanced failure theories as Continuum Damage Mechanics. For the time being this is still a field of research. Significant progress has been done within Airbus DS on the topic (see Ref. 3). However it still cannot be considered a standardized method, and few specialize engineering personnel is currently skilled to perform this kind of simulations with sufficient reliability. Future revisions of this documents shall treat this topic in detail as soon as judged reasonable.

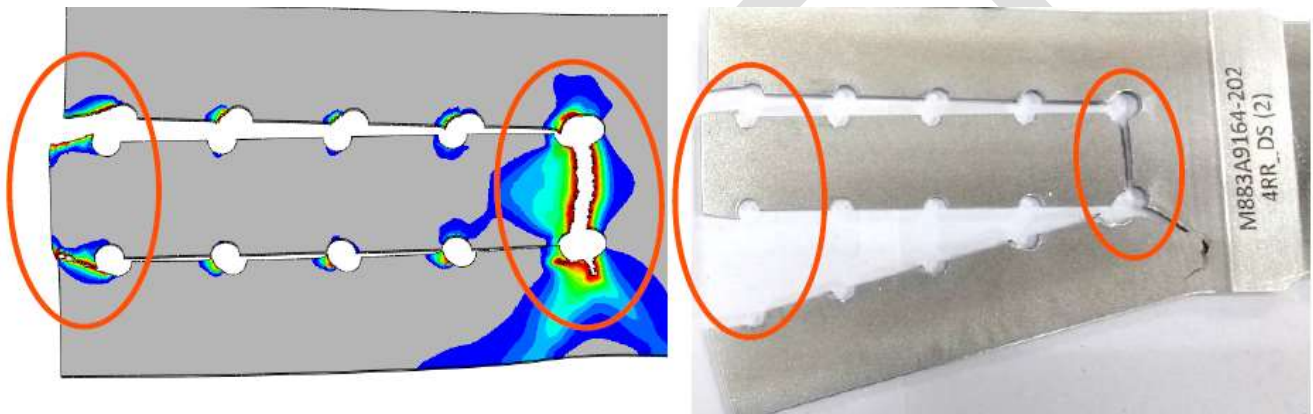


Figure 3. Advanced numerical simulation and test specimen after failure (Ref. 3)

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
F_{tu} , UTS	MPa	Ultimate tensile stress
TBC		To Be Confirmed

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[Airbus DS] NT-T-ADP-10003	Ed. B	Guía para la realización de Análisis de Estructuras mediante Modelos de Elementos Finitos de Detalle
Ref. 2	[Airbus] V53PR0906133	Is. 2.1	A350 Detailed FEM Modelling guidelines for Linear and Non Linear Analysis
Ref. 3	[Airbus DS] TEAS-NT-2010014	Is. A	Implementation of Continuum Damage Mechanics for Virtual Strength Testing

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
NASTRAN [MSC Software]	-	Software for Finite Element Analysis
Simulia ABAQUS [Dassault Systèmes]	-	Software for Finite Element Analysis

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.1 DD/01/2024	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

FITTINGS

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Methods & Optimization</i>	G. Baños DD/01/2024	Name DD/MM/AAAA
<i>Structural Analysis</i>	Name DD/MM/AAAA	F. Glock DD/01/2024

Table 3. Approval signatures table for last revision of this chapter